

# A 0D pulsatile model of aortic-arch banding

Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

July 25, 2026

# The problem

**Eberth et al. (2009):** a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with **similar mean pressure & flow but very different pulsatility**.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections  $\Rightarrow$  pulsatility index rises ( $PI \approx 3.11$ ).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse ( $PI \approx 1.65$ ); baseline CCA  $\approx 1.16$ .

**Goal:** a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

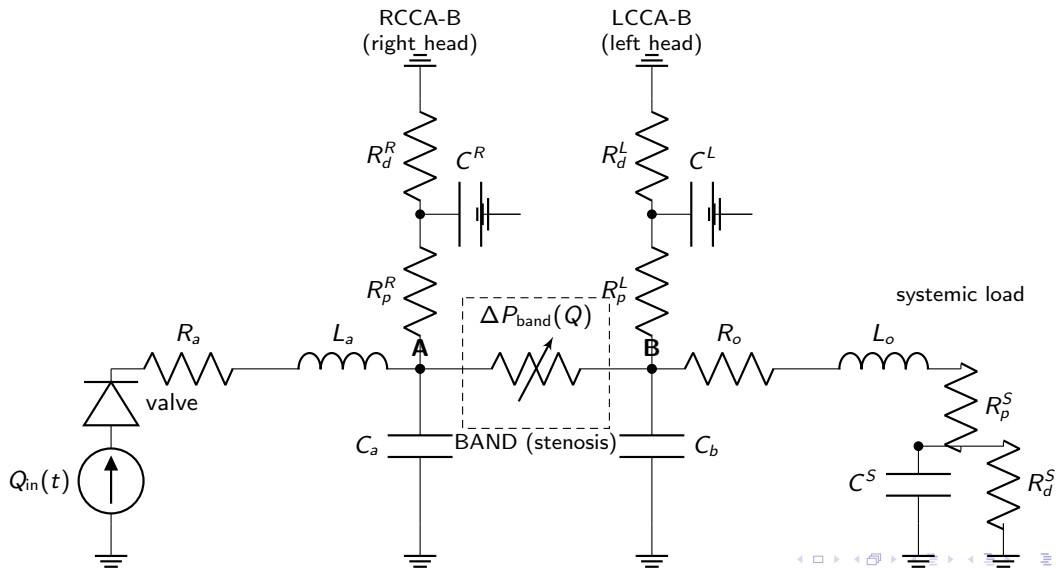
# The 0D model

Heart  $\rightarrow$  aortic valve  $\rightarrow$  ascending aorta  $\rightarrow$  **node A**  $\rightarrow$  band  $\Delta P(Q) \rightarrow$  **node B**  $\rightarrow$  descending aorta;  
a carotid RCR bed hangs off each node.

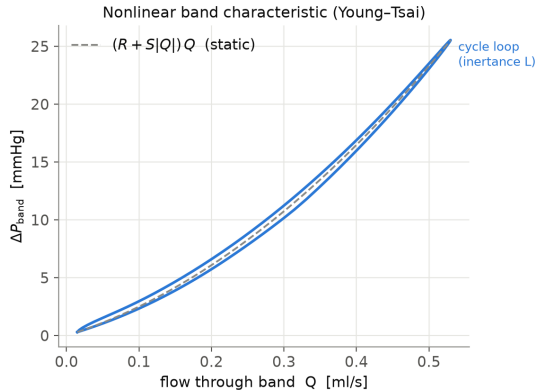
- Vessels: RCL BloodVessel blocks
- Band: nonlinear *Young-Tsai* stenosis
$$\Delta P = (R + S|Q|)Q + L \dot{Q}$$
- Terminal beds: 3-element RCR windkessels
- Valve: ValveTanh diode
- Parameters from mouse literature (geometry, blood, wall stiffness  $E$ )
- Every parameter tagged  
FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

# 0D circuit (electric analogy)



# The band is a nonlinear (Young–Tsai) element



$\Delta P$ – $Q$  across the band: the quadratic Bernoulli/turbulent term ( $S|Q|Q$ ) plus an inertance loop ( $L \dot{Q}$ ). This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

# Parameter roles: fixed, assumed, and the knob we tune

## Fixed (measured / literature)

- blood  $\rho = 1.06$ ,  $\mu = 3.5$  cP
- wall  $E$ : aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

⇒ all vessel  $R$ ,  $L$ ,  $C$  are *derived* from these via Eq. (1)–(3).

## Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split  $R_p:R_d = 10:90$ ; time const.  $\tau$
- distal pressure  $P_d = 0$
- valve  $R_{max}/R_{min}$ /steepness

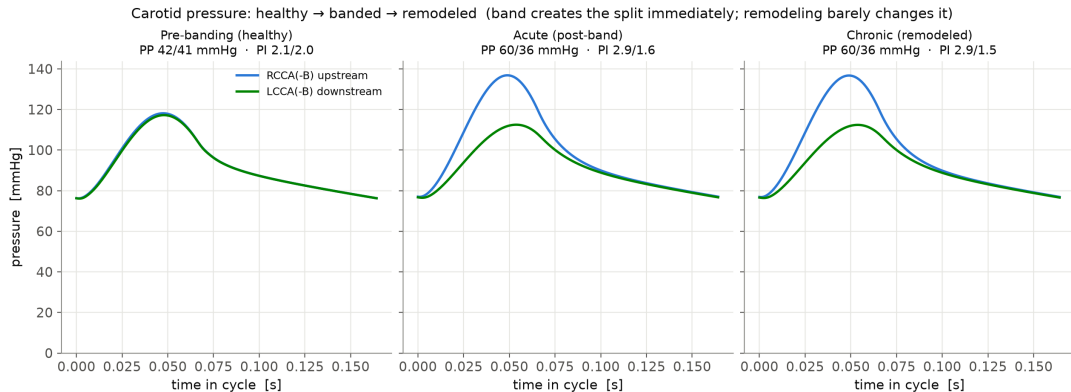
chosen by convention/estimate — could be refined, but not tuned to the data.

## Knob (calibrated to data)

- band stenosis\_coefficient  $S$
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

Everything else is fixed or derived — this is the only free knob fitted to the data.

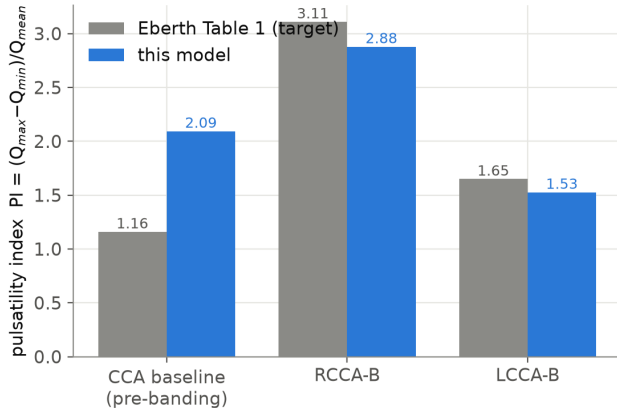
# Result: banding creates the split immediately



Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream  $\uparrow$ , downstream damped), and **barely changes** with wall remodeling  $\Rightarrow$  the split is set by band *position*; remodeling is the response, not the cause.

# Result: pulsatility vs. the paper

Pulsatility by location — calibrated model vs paper Table 1  
(stenosis calibrated to the PI ratio; both PIs within ~7%)



|           | Paper   | Model         |
|-----------|---------|---------------|
| RCCA-B PI | 3.11    | $\approx 2.9$ |
| LCCA-B PI | 1.65    | $\approx 1.5$ |
| PI ratio  | 1.89    | 1.89          |
| PP RCCA-B | 56      | 60            |
| MAP (R/L) | similar | 98/91         |

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.

## Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** ( $\approx 2.1$  vs. the paper's 1.16), though baseline *pulse pressure* matches ( $\approx 41$  vs. 42).
  - A parameter sweep shows every knob that lowers the baseline PI to 1.16 **also collapses the banded PI** — the passive bed is a linear pulsatility filter.
  - The paper's baseline→banded amplification is  $\sim 2.7\times$  ( $1.16\rightarrow 3.11$ ); a lumped 0D band delivers only  $\sim 1.5\times$ .
  - The missing  $\sim 2\times$  is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.
- ⇒ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D→1D boundary.

# Targets and knobs, by group

## Control group (CCA) — 4 targets

| target (measured)      | knob that sets it |
|------------------------|-------------------|
| $\bar{Q} = 0.016$ ml/s | carotid bed $R$   |
| MAP = 92 mmHg          | systemic $R$      |
| PP = 42 mmHg           | aortic $C$        |
| PI = 1.16              | carotid bed $C$   |

## Banded group (RCCA-B / LCCA-B) — 12 targets

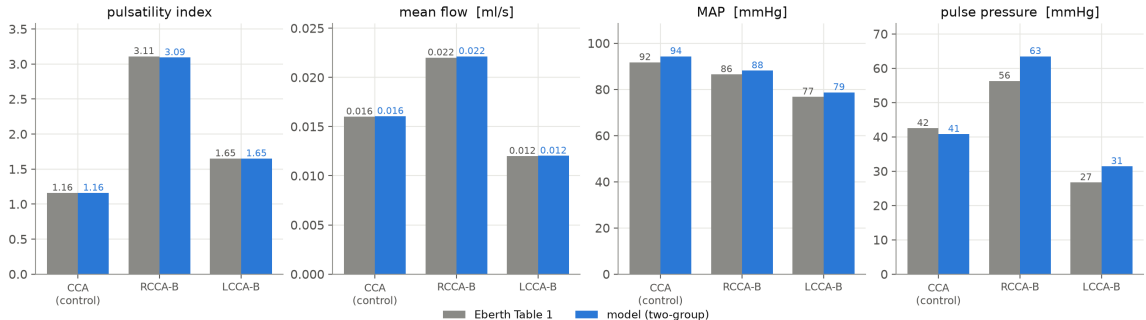
| target (measured)         | knob that sets it             |
|---------------------------|-------------------------------|
| $\bar{Q} = 0.022 / 0.012$ | RCCA / LCCA bed $R$           |
| MAP = 86 / 77             | band $S$ (A→B drop) + sys $R$ |
| PP = 56 / 27              | aortic $C_a, C_b$             |
| PI = 3.11 / 1.65          | RCCA / LCCA bed $C$           |

Bed resistances are *pinned* by ( $\bar{Q}$ , MAP); the actively-tuned knobs are the bed compliances ( $\rightarrow$  PI) and, for banded, the band  $S$  ( $\rightarrow$  the A→B pressure drop). Fixed for both groups: HR (7.17 / 6.09), cardiac output CO, carotid geometry, blood  $\mu, \rho$ , wall moduli  $E$ , ejection shape.

# Revised: two-group calibration (control vs. banded)

Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share  $R/C/L$ . Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured  $A \rightarrow B$  pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:

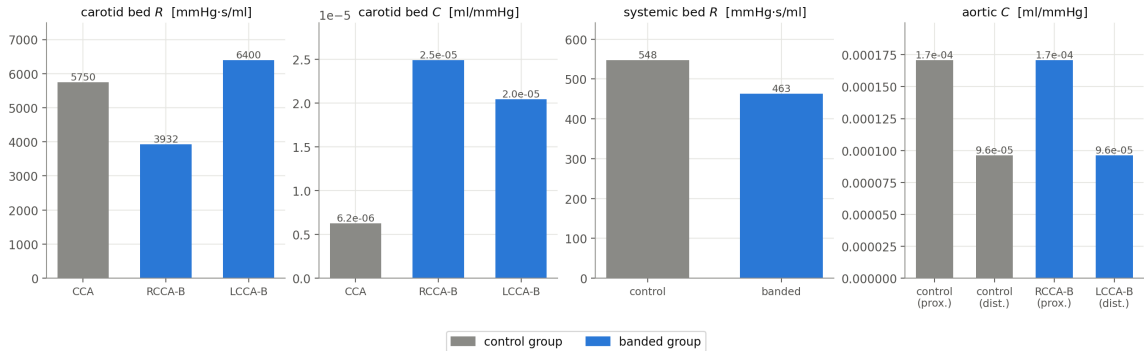
Two-group calibration vs Eberth Table 1 (control CCA + banded RCCA-B/LCCA-B)



DOF-balanced (12 targets / 12 knobs). Cost: bed compliance differs between groups (fitted, not measured) — descriptive, not a single predictive model.

# Settled calibration knobs: control vs. banded

Settled calibration knobs (R and C; band S excluded): control vs banded



Carotid bed  $R$  sets the flow split (RCCA-B low  $\rightarrow$  more flow); the banded carotid-bed *compliance* is  $\sim 3-4\times$  the control's — a **fitted** difference, not measured (opposite to what autoregulation would do). Systemic  $R$  is essentially unchanged; aortic  $C$  (both the proximal and distal segment, either side of the band) is **not tuned in either group** — same 0.64/0.36 split of the measured central aortic compliance for control and banded alike.

# Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.